Czech Clinical Named Entity Recognition for MultiClinAI: Morphological Augmentation, CRF Decoding, and Ensemble Voting

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Abstract

We present our system for Czech clinical Named Entity Recognition (NER) submitted to the MultiClinAI shared task, targeting three entity types: disease, symptom, and procedure. Czech poses particular challenges as a morphologically rich, inflectional language with limited annotated clinical resources. Our approach fine-tunes multilingual (XLM-RoBERTa-base) and Czech-specific (RobeCzech) transformer models, optionally combined with a Conditional Random Field (CRF) decoding layer. The most impactful contribution is a morphological synonym augmentation strategy that generates additional training documents by replacing rare entity surface forms with morphological variants from a curated Czech medical synonym dictionary, yielding a consistent +4% F1 improvement across all entity types. We further improve performance through weighted ensemble voting across multiple model configurations and identify a tokenizer artifact in XLM-RoBERTa's SentencePiece byte-fallback that produced spurious short entity spans, whose resolution brought a substantial gain on the test set. Our best system achieves strict F1 scores of 0.691 (disease), 0.662 (procedure), and 0.634 (symptom) on the official MultiClinNER Czech test set.

1 Introduction

Clinical documents encode a wealth of medical knowledge, but extracting structured information from free-text narratives requires dedicated natural language processing (NLP) systems. Named Entity Recognition (NER) is a foundational step in clinical information extraction, enabling downstream tasks such as diagnosis coding, pharmacovigilance, and population health analysis. The MultiClinAI shared task (Lima López et al., 2026) challenges participants to build NER systems for seven languages, extracting three entity types (DISEASE, SYMPTOM, PROCEDURE) from clinical case reports.

Czech presents distinct challenges compared to other task languages. As a highly inflectional Slavic language, the same medical concept may appear in many morphological forms depending on grammatical case, number, and gender. For example, the disease *hypertenze* (hypertension) appears as *hypertenzí* (genitive), *hypertenzi* (dative/accusative), or *hypertenzí* (instrumental) in natural text. Standard surface-form matching and even pretrained language models may not generalise well across all inflected forms if they are underrepresented in training data. Furthermore, the MultiClinNER Czech training set comprises approximately 1,006 documents per entity type, which is relatively small for fine-tuning large transformer models reliably.

This paper describes our system for the Czech track of MultiClinNER (Subtask 1: entity detection). Our main contributions are:

- A **morphological synonym augmentation** strategy targeting rare entity surface forms that generates 2,161–2,271 additional training documents per entity type, improving F1 by approximately 4 percentage points.
- A systematic comparison of multilingual (XLM-RoBERTa-base) vs. Czech-specific (RobeCzech) transformer models, with and without CRF decoding.
- A **sentence-level training** procedure with Czech-aware abbreviation detection that improves training efficiency while maintaining entity/context balance.
- Identification and correction of a **SentencePiece tokenizer artifact** that caused contradictory BIO label sequences and spurious short-span predictions.
- A **weighted ensemble voting** strategy combining fine-tuned models with a GPT-4.1-mini

zero-shot model for complementary recall.

2 Related Work

Transformer-based NER. Transformer models pretrained on large corpora (Devlin et al., 2019) have become the standard for sequence labeling tasks. Token classification with a linear head over contextual embeddings achieves strong results on many NER benchmarks. Multilingual models such as mBERT and XLM-RoBERTa (Conneau et al., 2020) enable transfer across languages, which is especially valuable in low-resource settings.

CRF decoding. Conditional Random Fields (CRFs) (Lafferty et al., 2001) model label transition dependencies explicitly, enforcing globally consistent BIO sequences via Viterbi decoding. Combined with transformer encoders (Ma and Hovy, 2016), CRF heads prevent invalid label sequences (e.g., `I-X` following `O`) and can improve precision in span-boundary detection.

Clinical and biomedical NER. Clinical NER has benefited substantially from domain-adaptive pretraining. Models pretrained on biomedical text such as BioBERT (Lee et al., 2020) and clinical corpora outperform general-domain models on tasks like disease and drug extraction. For Spanish clinical NER, CLIN-X-ES (Gutiérrez-Fandiño et al., 2022) demonstrated the value of language-and-domain-specific pretraining. The Czech language has seen less work in the clinical domain; RobeCzech (Straka and Straková, 2021) is the primary Czech monolingual language model, pretrained on large web and news corpora.

Data augmentation for NER. The scarcity of annotated clinical data motivates augmentation approaches. Wei and Zou (2019) propose simple operations (synonym replacement, insertion, deletion, swap) that improve text classification. For NER specifically, synonym substitution must preserve entity boundaries and annotation offsets. Dai and Adel (2020) analyse various NER augmentation strategies and find that synonym replacement is among the most effective. We extend this approach with morphologically varied synonyms appropriate for an inflectional language.

LLM-based NER. Large Language Models (LLMs) applied zero-shot or few-shot to NER have shown competitive performance in general domains (Wang et al., 2023), but tend to underperform supervised models on specialised clinical terminology. Their high precision on rare entities makes them useful as diversity sources in ensemble systems.

3 Data

3.1 MultiClinNER Czech Dataset

The MultiClinNER Czech training data is derived from the DisTEMIST, SympTEMIST, and MedProcNER corpora translated and annotation-projected to Czech (Lima López et al., 2026). Documents are clinical case reports annotated in BRAT standoff format with character-offset entity spans. Each entity type (disease, symptom, procedure) has a separate set of documents; per the task guidelines, texts are not shared across entity types due to translation revision differences, precluding multi-label training.

Table 1 summarises the dataset statistics for Czech.

Entity	Documents		Entities	
	Train	Dev	Train	Dev
Disease	1,006	252	21,083	4,710
Symptom	1,006	252	22,525	5,281
Procedure	1,006	252	22,438	5,063

Table 1: MultiClinNER Czech dataset statistics (entity counts from gold `.ann` files).

3.2 Morphological Synonym Augmentation

To address the morphological richness of Czech and the limited training set size, we constructed a Czech medical synonym dictionary mapping entity surface forms to morphological variants. The dictionary was built by combining: (1) manually curated morphological paradigms for Czech medical terms, and (2) Wikidata concept labels for Czech medical entities. The final dictionary contains 1,399 synonym entries spanning 887 distinct morphological families.

Augmented documents are generated as follows. For each entity surface form in the training set that occurs at most 5 times (rare entities), we look up all available synonyms and generate one new document per synonym by replacing the entity with the synonym and adjusting all downstream character offsets accordingly. Entities with more than 5 occurrences are considered well-represented and excluded to avoid over-amplifying already common forms. This process yields 2,161 new disease documents, 2,271 new symptom documents, and

2,271 new procedure documents, more than tripling the training set size.

3.3 LLM Paraphrase Augmentation

We additionally generated paraphrases using GPT-4.1-mini following the BioSynNER methodology, where the model is prompted to rewrite clinical sentences while preserving entity spans inline. This produced supplementary paraphrase data at approximately $3.30 per entity type. The paraphrase augmentation provided a small additional improvement (0.002–0.004 F1 on development set) beyond morphological augmentation.

3.4 External Data (Negative Result)

We translated the NCBI Disease corpus (792 EN documents) to Czech via MarianMT with marker-based entity alignment (83% marker survival rate), but training on this silver data alone (F1=0.677) or combined with morphological augmentation (F1=0.689) both underperformed morphological augmentation alone (F1=0.710), indicating that MT projection noise outweighs the coverage benefit. DisTEMIST/SympTEMIST were excluded as data leakage (they are the source documents from which the competition training set was derived).

4 Methods

4.1 Base Models

We evaluate two pretrained transformer models for Czech:

XLM-RoBERTa-base (Conneau et al., 2020) (`FacebookAI/xlm-roberta-base`): A multilingual model pretrained on 2.5TB of filtered CommonCrawl data across 100 languages using masked language modelling. Despite not being specialised for Czech or clinical text, its large multilingual pretraining vocabulary handles Czech morphological forms well through SentencePiece subword tokenisation.

RobeCzech (Straka and Straková, 2021) (`ufal/robeczech-base`): A Czech-specific RoBERTa model pretrained on Czech web text, news, and Wikipedia. Its Czech-optimised vocabulary produces fewer subword splits per token for Czech words, potentially yielding better token-level entity alignment.

4.2 Model Architecture

Both models are fine-tuned for token classification in a BIO (Begin-Inside-Outside) tagging scheme, with one model trained per entity type. Separate models for each entity type are used because the task provides separate annotated text versions per entity.

The architecture consists of:

1. The pretrained transformer encoder producing contextual token representations.
2. A dropout layer (rate 0.1) for regularisation.
3. A linear projection layer mapping hidden states to the label space (7 labels: `B-TYPE`, `I-TYPE`, `O` for each entity type, plus padding label).
4. Optionally, a **Conditional Random Field (CRF)** layer replacing the standard argmax decoder with Viterbi-decoded label sequences.

During training with CRF, we minimise the negative log-likelihood of the gold label sequence. At inference, Viterbi decoding finds the globally optimal BIO sequence, preventing invalid transitions (e.g., `I-X` after `O`).

Special tokens (`[CLS]`, `[SEP]`) and non-first subword tokens receive label -100 and are excluded from both CRF and cross-entropy loss computation. Only the first subword token of each word receives a valid label; subsequent subword tokens inherit the outside label for loss purposes.

4.3 Tokenizer Artifact Fix

During development, we identified a critical artifact in XLM-RoBERTa's SentencePiece byte-fallback tokenisation: for certain byte sequences (typically non-ASCII characters in rare contexts), the tokeniser emits two consecutive tokens at identical character start offsets. This occurred approximately 4 times per document in the Czech clinical corpus.

When the standard label alignment algorithm processes these overlapping tokens, it assigns a `B-TYPE` label to both tokens at the same position, creating a `B`→`B` transition in the label sequence that contradicts BIO semantics. During training, this introduces inconsistent supervision; during inference, it causes the entity reconstruction to close one span and immediately open another, often producing 1–2 character spans corresponding to subword fragments rather than meaningful clinical terms.

We resolve this by:

- **Training**: Overlapping tokens (same start offset as the preceding token) receive label -100 and are excluded from the loss.

- **Inference**: When an overlapping token is encountered, the entity span is extended to the end of the overlapping token rather than split.
- **Post-filtering**: All predicted spans of 1 character are discarded. Spans of exactly 2 characters are retained only if they are fully uppercase (valid medical abbreviations such as *DM*, *HT*).

Table 4 shows the impact of this fix on the test set.

4.4 Training Setup

We train all models with the following hyperparameters: learning rate 2×10^{-5} (AdamW), weight decay 0.01, linear warmup for 10% of steps, batch size 16, maximum sequence length 512 tokens, and early stopping with patience 3 (monitoring development F1). Final submission models are retrained on the combined train+dev split for a fixed 10 epochs (no early stopping).

Sentence-level training. Rather than processing full documents (which may have variable lengths), we split each document into sentences and train on sentence-level examples. This increases the effective batch diversity and entity-to-token ratio per batch. Sentence boundaries are detected using punctuation followed by whitespace and an uppercase letter, with a curated list of Czech abbreviations (e.g., *mudr.*, *např.*, *tj.*) that suppress false splits, and a decimal-number check to avoid splitting on numerical values. We retain pure-`O` sentences in training to prevent the model from developing a recall bias (empirically, removing them drops precision to 0.418).

Oversampling. Documents containing at least one entity annotation are oversampled once ($1\times$) in the training set before merging with augmented data. This further increases the proportion of entity-containing examples without over-weighting augmented data.

4.5 Ensemble Method

We combine multiple independently trained model checkpoints through span-level voting. Each model predicts a set of (start, end, type) entity spans; the ensemble aggregates these predictions using one of several strategies explored in a development-set sweep:

- **Threshold-K**: Accept a span if at least K models predict it.
- **Weighted voting**: Each model is assigned a weight proportional to its development F1 score. A span is accepted if the sum of voting model weights exceeds a threshold t (swept over $[0.3, 0.8]$).
- **Two-tier**: A core agreement layer (at least K models) plus an additional trust layer accepting unique predictions from high-precision models.

For disease, the best development-set ensemble combines four models: XLM-R+morph+oversample, RobeCzech+morph+paraphrase, RobeCzech+morph, and GPT-4.1-mini (zero-shot). Despite GPT-4.1-mini's low individual F1 (0.384), its predictions have 65.8% complementarity with the fine-tuned models: it finds 224 entities missed by all fine-tuned models, including rare terms, complex multi-word spans, and abbreviations. This boosts ensemble recall from 0.719 to 0.730 with weighted voting at $t = 0.4$.

5 Experiments and Results

5.1 Ablation Study (Development Set)

Table 2 presents the development-set ablation for disease NER, comparing the effect of model choice, CRF decoding, and augmentation strategy. Results for symptom and procedure show consistent trends.

Model	Aug.	Head	Extra	F1
XLM-R-base	None	Softmax	–	0.657
XLM-R-base	Morph	Softmax	–	0.710
XLM-R-base	Morph	CRF	–	0.712
XLM-R-base	Morph	Softmax	OS×1	0.710
XLM-R-base	Morph+Para	Softmax	–	0.704
RobeCzech	Morph	Softmax	–	0.698
RobeCzech	Morph+Para	Softmax	–	**0.714**

Table 2: Development set disease F1 ablation. Aug. = augmentation (Morph = morphological synonyms, Para = LLM paraphrase); Extra: OS×1 = 1× entity-doc oversampling. XLM-R-base = `FacebookAI/xlm-roberta-base`; RobeCzech = `ufal/robeczech-base`.

Key findings from the ablation:

Morphological augmentation is the single most effective intervention, adding approximately 4 percentage points F1 over the non-augmented baseline (0.657 → 0.710–0.712).

CRF decoding provides a small but consistent improvement over softmax for the same model and

augmentation (XLM-R + morph: 0.710 → 0.712), by enforcing valid BIO transitions during decoding.

RobeCzech vs. XLM-RoBERTa: Despite being Czech-specific, RobeCzech slightly underperforms XLM-RoBERTa with morphological augmentation alone (0.698 vs. 0.710). However, RobeCzech benefits more from LLM paraphrase augmentation, reaching 0.714 to become the best single model—suggesting its Czech-optimised tokenisation extracts more value from paraphrased clinical text.

LLM paraphrases add a marginal +0.004 improvement beyond morphological augmentation alone, indicating diminishing returns when the morphological variants already cover the most impactful diversity.

Oversampling entity-containing documents (OS×1) yields no additional gain beyond morphological augmentation, though it was beneficial without augmentation.

5.2 Ensemble Results (Development Set)

System	Strategy	P	R	F1
4-model	Weighted t=0.4	0.747	0.730	**0.738**
	Threshold K=2	0.743	0.729	0.736
	Majority K=3	0.820	0.629	0.712
	Union K=1	0.694	0.731	0.712
3-model	Threshold K=2	0.753	0.719	0.736
Best single	–	0.718	0.711	0.714

Table 3: Disease development set ensemble results (dev set, 4,710 gold entities). 4-model: XLM-R+morph+OS, RobeCzech+morph+para, RobeCzech+morph, GPT-4.1-mini. 3-model: same without GPT-4.1-mini. Best single: RobeCzech+morph+para.

The weighted voting ensemble with t=0.4 achieves F1=0.738, outperforming the best single model by 2.4 percentage points. Notably, including the GPT-4.1-mini zero-shot model increases recall (0.719 → 0.730) without substantially hurting precision (0.753 → 0.747), since most of its unique predictions are genuine entities missed by the fine-tuned models. The 3-model variant (excluding GPT) achieves the same F1 (0.736) through higher precision (0.753 vs. 0.747), providing a useful precision-recall trade-off.

For procedure and symptom, two-model ensembles (XLM-R+CRF and RobeCzech+OS×1) were used since fewer model variants were trained for these entity types.

Entity	System	P	R	F1
Disease	Ensemble (pre-fix)	0.565	0.674	0.615
	Ensemble (post-fix)	0.708	0.675	**0.691**
Procedure	Ensemble (pre-fix)	0.617	0.685	0.649
	Ensemble (post-fix)	0.634	0.692	**0.662**
Symptom	Ensemble (pre-fix)	0.564	0.629	0.595
	Ensemble (post-fix)	0.695	0.582	**0.634**

Table 4: Test set impact of the SentencePiece tokenizer artifact fix. Pre-fix results from official evaluation of pre-bugfix submissions; post-fix from bugfix submissions.

5.3 Bug Fix Impact

The SentencePiece fix brings substantial improvements, particularly in precision (disease: 0.565 → 0.708, symptom: 0.564 → 0.695). This confirms that overlapping token positions were generating large numbers of spurious short entity spans, inflating false positives.

5.4 Test Set Results

Entity	System	P	R	F1
Disease	Ensemble (4-model)	**0.708**	**0.675**	**0.691**
	XLM-R CRF+morph	0.651	0.638	0.645
	XLM-R+morph+OS	0.650	0.639	0.644
Procedure	Ensemble (2-model)	0.634	**0.692**	**0.662**
	XLM-R CRF+morph	**0.668**	0.655	0.662
Symptom	Ensemble (2-model)	**0.695**	0.582	**0.634**
	XLM-R CRF+morph	0.612	0.581	0.596

Table 5: Official MultiClinNER Czech test set results (strict span matching, post-bugfix). Ensemble models: Disease = XLM-R×2 + RobeCzech×2; Procedure/Symptom = XLM-R CRF + RobeCzech+OS.

The ensemble consistently outperforms individual models across all entity types. The largest gain from ensembling is seen for symptom (+0.038 F1), where the individual CRF model's lower recall (0.581) is partially recovered by the complementary RobeCzech model. For procedure, both ensemble and CRF achieve identical F1 (0.662), with the ensemble recovering higher recall at the cost of lower precision.

Comparing entity types, disease achieves the highest test F1 (0.691), followed by procedure (0.662) and symptom (0.634). The lower symptom performance likely reflects the inherently more ambiguous and context-dependent nature of symptom mentions, which can overlap with disease and procedure terminology.

5.5 Negative Results

A CZ→EN→NER pipeline (MarianMT translation + Stanza i2b2 NER + awesome-align projection back to Czech offsets) achieved only F1=0.107 on the disease development set due to English NER over-prediction (∼8K spans per entity type) and span fragmentation during translation, and was not used. Augmenting with Wikidata true synonyms (F1=0.693) slightly underperformed morphological-only augmentation (F1=0.697), as concept-level lexical substitution diverged from the clinical register and introduced noise. XLM-RoBERTa-large yielded only +0.4% F1 over the base model at twice the training cost and was excluded from the final system.

6 Error Analysis

6.1 GPT-4.1-mini Complementarity

To understand the sources of ensemble gain from GPT-4.1-mini, we analyse its predictions on the development set alongside the best fine-tuned model (RobeCzech+morph+para).

Metric	RobeCzech	GPT-4.1-mini
Individual F1	0.714	0.384
Entities found	3,350 (71.1%)	1,447 (30.7%)
False positives	1,318	1,431
Unique correct	2,127	**224**
Union recall	75.9% (+4.8% vs. single)	
Complementarity	65.8%	

Table 6: Comparison of RobeCzech+morph+para and GPT-4.1-mini on the Czech disease development set (4,710 gold entities). Percentages are out of 4,710. Complementarity: share of GPT's correct finds that RobeCzech misses.

GPT-4.1-mini uniquely recovers 224 disease entities missed by all fine-tuned models. Analysis of these unique finds reveals three main categories:

- **Rare and complex multi-word terms** (e.g., *mezenchymální tumor*, *juxtaglomerulární buněčný tumor*): fine-tuned models tend to split or partially predict these spans.
- **Medical abbreviations** (e.g., *PKU*): undertrained in the limited Czech clinical corpus but known to GPT from broader pretraining.
- **Contextual or implicit disease mentions**: where the model must infer disease status from surrounding sentence context.

6.2 Remaining Error Patterns

Approximately 24% of gold entities in the disease development set are missed by all models including GPT-4.1-mini. Manual inspection reveals:

- **Very long entity spans**: Multi-clause noun phrases where span boundaries are ambiguous even for annotators.
- **Highly inflected rare forms**: Morphological forms with fewer than 1–2 training examples even after augmentation.
- **Negated or hypothetical mentions**: Entities appearing in constructions such as *bez* (without), *vyloučit* (to rule out), which our models treat as positive examples regardless of assertion status.
- **English-Czech code-mixing**: Medical abbreviations written in English in Czech clinical text (e.g., *COPD*, *GERD*).

False positives are predominantly caused by **adjacent medical terms** being merged into a single longer span and **general medical vocabulary** (anatomical terms, test names) being incorrectly classified as disease mentions.

7 Conclusion

We presented a Czech clinical NER system for the MultiClinAI MultiClinNER shared task, achieving strict F1 scores of 0.691, 0.662, and 0.634 for disease, procedure, and symptom respectively. Our main finding is that **morphological synonym augmentation** is the most impactful single intervention for Czech clinical NER, providing a consistent +4% F1 improvement by generating morphologically varied training documents for rare entity surface forms. CRF decoding offers small additional gains by enforcing BIO sequence validity. Ensemble voting across diverse model configurations provides further gains, with GPT-4.1-mini improving recall on rare and complex entity spans.

We also highlight the importance of **careful tokenizer handling**: the SentencePiece byte-fallback artifact in XLM-RoBERTa caused significant precision degradation (precision dropped to 0.565 for disease) that was recoverable only after diagnosing and fixing overlapping token alignment. We recommend that practitioners explicitly handle this edge case when applying XLM-RoBERTa to non-ASCII clinical text.

Limitations. Our system does not address assertion status (negated or hypothetical entities), which may cause false positives in clinical text. The morphological synonym dictionary was constructed semi-automatically and may contain errors. Results were obtained on a single test set and may not generalise across clinical specialities.

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